

WILLIAM NASH

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SUMMARY

Head of Engineering with 10+ years in distributed systems and scientific computing, from LHC analysis at CERN to machine learning on petabyte-scale satellite data for carbon monitoring at Chloris. Climate-motivated. Breakthrough Prize in Fundamental Physics (CMS Collaboration, 2025).

PROFESSIONAL EXPERIENCE

Chloris Geospatial

Head of Engineering

May 2022 – Present

Boston, MA

- Built a petabyte-scale geospatial data platform as one of two engineers on a team growing to 10; its Dask workloads on thousands of AWS workers produce Chloris's commercial product, global carbon-stock maps spanning 25 years.
- Designed the engine behind Chloris's customer-facing carbon statistics for 72 countries: it aggregates arbitrary regions across the global Sentinel-2 land grid (~20,000 tiles in differing projections) without reprojecting pixels.
- Built a customer data API and hosted MCP server on Kubernetes (EKS) that let LLM agents query carbon-stock data by polygon; in internal release ahead of customer launch.
- Engineered the production ML pipeline behind Chloris's global carbon-stock maps, from satellite feature generation through model training to continental-scale distributed inference.
- Now re-architecting it as a next-generation platform: Dagster-orchestrated builds of sharded Zarr cubes, GPU inference for PyTorch models on optical and radar time series, and versioned products in a pgSTAC catalog.
- Made spatially correlated confidence intervals on carbon aggregates tractable at continental scale: approximated an $O(N^2)$ error-covariance sum over hundreds of millions of pixels with a banded stencil, distributed with Dask.
- Sped up pipeline kernels 3–8x with Numba JIT compilation, and replaced an exponential-time changepoint algorithm with a greedy $O(N^3)$ variant that matches the exact result on 99.2% of pixels.

UCLA Physics Department / CERN

PhD Candidate, then Postdoctoral Researcher

July 2016 – March 2022

Los Angeles, CA

- Designed a C++ pattern-recognition algorithm for the real-time muon trigger of the CMS detector; its lookup tables were deployed to FPGA firmware at zero added latency, improving position resolution 2x in LHC Run 3.
- Built custom reconstruction and data-driven background estimation, in C++ and distributed Python (Dask) on the CERN grid, for a long-lived-particle search over quadrillions of proton collisions; its limits were world-leading.
- Developed a data-driven framework to quantify systematic uncertainties in displaced-muon reconstruction, calibrating trigger efficiencies with cosmic-ray data.

Mevion Medical Systems

Software Engineer (Physics and Algorithms)

June 2014 – July 2016

Littleton, MA

- Developed real-time algorithms that modulate proton beam position on a 250 MeV synchrocyclotron used for cancer treatment (Mevion now equips ~56% of US single-room proton therapy centers).
- Built a GEANT4 particle-transport simulation batch farm on AWS for radiation-field modeling and verification.

EDUCATION

UCLA, Los Angeles, CA

PhD in Physics

March 2022

Boston University, Boston, MA

BA in Physics (*cum laude*)

May 2014

TECHNICAL SKILLS

Languages Python, C++, JavaScript, Unix Shell

Scientific GDAL, Xarray, Zarr, STAC, COG, Numba, GEANT4, Monte Carlo, ROOT, HTCondor

ML/Data PyTorch, LightGBM, scikit-learn, Dask, Dagster, NumPy, pandas, Parquet

Cloud/Infra AWS (Batch, Step Functions, S3), Kubernetes (EKS), Docker, Terraform, GitLab CI, SQL

AI Tooling Claude Code, Anthropic SDK, n8n, MCP servers, LLM-orchestrated workflows

AWARDS & PUBLICATIONS

Breakthrough Prize in Fundamental Physics, awarded to the LHC collaborations (CMS member)

2025

AWS Certified Machine Learning Specialty

2023

CMS Collaboration, "Search for long-lived particles decaying to a pair of muons in pp collisions at $\sqrt{s} = 13$ TeV", JHEP05(2023)228.